

1. Notation & General Map

N	total units (e.g. 6 plots)
N_T, N_C	sizes of treatment / control arms
$ W^+ $	# legal assignment vectors
$P(W)$	$= 1/ W^+ $ under complete rand.
$Y_i(w)$	potential outcome of unit i at w
τ_i	$Y_i(1) - Y_i(0)$, unit-level effect
τ	$\frac{1}{N} \sum \tau_i$, average causal effect
$\hat{\tau}$	estimator of τ from observed data
Y_i^{obs}	observed outcome of i under W_i
$\bar{Y}_T^{obs}, \bar{Y}_C^{obs}$	observed arm means
s_j^2	within-arm sample variance $\div (N_j - 1)$
S_τ^2	finite-pop variance of τ_i (unobs.)

Two inference styles — don't mix!

Fisherian: sharp null $\tau_i = 0 \forall i$; Y^{obs} fixed; reference set comes from design; compute p -value.

Neymanian: estimate $\hat{\tau}$ and its variance; build $\hat{\tau} \pm 2 \cdot SE$ CI (95% uses **2**, not 1.96 – STAT 140 convention).

2. CRE (Completely Randomized)

Setup

N units split into N_T treated and N_C controls by simple random assignment.

$$|W^+| = \binom{N}{N_T}, \quad P(W_i=1) = \frac{N_T}{N} = \frac{1}{2} \text{ if balanced.}$$

Neymanian steps

- Arm means: $\bar{Y}_T^{obs} = \frac{1}{N_T} \sum_{W_i=1} Y_i^{obs}$ (same for C).
- Estimator: $\hat{\tau} = \bar{Y}_T^{obs} - \bar{Y}_C^{obs}$.
- Sample variances: $s_j^2 = \frac{1}{N_j-1} \sum_{i \in j} (Y_i^{obs} - \bar{Y}_j^{obs})^2$.
- Conservative** variance estimate:

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{s_1^2}{N_T} + \frac{s_0^2}{N_C}.$$

- 95% CI: $\hat{\tau} \pm 2 \cdot \sqrt{\widehat{\text{Var}}(\hat{\tau})}$.

Fisherian steps

Sharp null $H_0 : Y_i(1) = Y_i(0) \forall i$. Under H_0 all N numbers are fixed; only labels move.

Test stat: $T_{dm} = \bar{Y}_T - \bar{Y}_C$.

$$p = \frac{\#\{W \in W^+ : |T(W)| \geq |T^{obs}|\}}{|W^+|}.$$

If $|W^+|$ big (e.g. $\binom{30}{15} \approx 1.5 \times 10^8$) $\Rightarrow$ Monte Carlo $B=10,000$ draws.

Why variance is conservative

True variance $= \frac{s_1^2}{N_T} + \frac{s_0^2}{N_C} - \frac{S_\tau^2}{N}$. Last term $S_\tau^2/N \geq 0$ but needs *both* potential outcomes per unit – **unobservable**. Dropping it inflates the estimate $\Rightarrow$ upper bound (still valid for 95% CI, just wider).

Mini worked example ($N=6, N_T=N_C=3$)

Observed $Y_T^{obs} = \{30, 32, 34\}, Y_C^{obs} = \{22, 21, 24\}$. $\hat{\tau} = 32 - 22.33 = 9.67$; $s_1^2 = 4, s_0^2 \approx 2.33$; $\widehat{\text{Var}} \approx 2.11, SE \approx 1.45$; CI $= (6.76, 12.58)$.

3. Matched Pair Design

Setup

N units in $P = N/2$ pairs (matched on pre-treatment covariate). Within each pair, flip a coin: one unit $\rightarrow T$, one $\rightarrow C$.

$$|W^+| = 2^P, \quad P(W) = 1/2^P.$$

Neymanian steps

- Within-pair difference: $d_p = Y_{T,p}^{obs} - Y_{C,p}^{obs}$.
- Estimator: $\hat{\tau} = \bar{d} = \frac{1}{P} \sum d_p$.
- Sample variance of differences: $s_d^2 = \frac{1}{P-1} \sum (d_p - \bar{d})^2$.
- Exact** variance estimate:

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{s_d^2}{P}.$$

- 95% CI: $\bar{d} \pm 2\sqrt{s_d^2/P}$.

Fisherian steps

Under sharp null each d_p 's sign is a fair coin flip (pair-internal rand.). Test stat: $T = \bar{d}$. Reference set: 2^P sign patterns $\epsilon \in \{-1, +1\}^P$, $d_p^* = \epsilon_p |d_p|$. $p = \#\{|\bar{d}^*| \geq |\bar{d}^{obs}|\} / 2^P$.

Why variance is exact (not conservative)

d_p is **directly observable** for every pair – no hidden S_τ^2 in the formula. So s_d^2/P is an unbiased estimate of the true $\text{Var}(\hat{\tau})$, not an upper bound.

Key trap

Reference set is 2^P , **NOT** $\binom{N}{N/2}$. The $\binom{N}{N/2}$ would allow swapping fish across pairs, violating the design.

Mini worked example ($P=4$ pairs)

Observed $d = \{3, 1, 5, 3\}$. $\bar{d} = 3$; devs $\{0, -2, 2, 0\} \Rightarrow s_d^2 = 8/3 \approx 2.67$; $\widehat{\text{Var}} = s_d^2/4 = 2/3 \approx 0.67, SE \approx 0.82$; CI $= 3 \pm 1.63 = (1.37, 4.63)$. Fisherian: $T^{obs} = 3$; of the $2^4 = 16$ sign patterns, only $(+, +, +, +)$ and $(-, -, -, -)$ give $|\bar{d}^*| \geq 3 \Rightarrow p = 2/16 = 0.125$.

4. Multi-Treatment CRE

Setup

N units allocated to J arms with sizes $N_1, \dots, N_J, \sum N_j =$

N .

$$|W^+| = \frac{N!}{\prod_j N_j!}.$$

ANOVA decomposition (formulas)

$\bar{Y} = \frac{1}{N} \sum_i Y_i^{obs}$ (grand mean).

$SS_{Treat} = \sum_{j=1}^J N_j (\bar{Y}_j - \bar{Y})^2$ (between-arm).

$SS_{Res} = \sum_j \sum_{i \in j} (Y_i^{obs} - \bar{Y}_j)^2$ (within-arm).

$SS_{Total} = \sum_i (Y_i^{obs} - \bar{Y})^2 = SS_{Treat} + SS_{Res}$.

Generic ANOVA table

Source	df	SS	MS	F
Treatments	$J-1$	SS_{Treat}	$\frac{SS_{Treat}}{J-1}$	$\frac{MS_{Treat}}{MS_{Res}}$
Residuals	$N-J$	SS_{Res}	$\frac{SS_{Res}}{N-J}$	–
Total	$N-1$	SS_{Total}	–	–

Intuition for F

Under sharp null, $MS_{Treat} \approx MS_{Res} \Rightarrow F \approx 1$. Large $F \gg 1 \Rightarrow$ reject. F -test is **one-sided** (large values only).

Pairwise Neymanian

$\hat{\tau}(j, j') = \bar{Y}_j - \bar{Y}_{j'}$; conservative var $= s_j^2/N_j + s_{j'}^2/N_{j'}$. **Conservative** because exact variance contains $S_{jj'}^2/N$ needing both potential outcomes per unit.

Multiple testing caution

$\binom{J}{2}$ pairwise tests $\Rightarrow$ family-wise error $\approx 1 - (1 - \alpha)^{\binom{J}{2}}$. Gate with the F -test first, then do pairwise.

Fisherian for F

Under sharp null, permute unit labels across arms (total $|W^+|$ permutations). Recompute F^{rand} each time. $p = \#\{F^{rand} \geq F^{obs}\} / |W^+|$ (one-sided).

Mini worked example ($J=3, N=9$)

Arms: std $\{5, 7, 6\} \rightarrow \bar{Y}_1 = 6$; low $\{8, 9, 7\} \rightarrow \bar{Y}_2 = 8$; high $\{12, 11, 10\} \rightarrow \bar{Y}_3 = 11$. Grand $\bar{Y} = 25/3$.

$SS_{Treat} = 3[(6-25/3)^2 + (8-25/3)^2 + (11-25/3)^2] = 3(49+1+64)/9 = 38$.

$SS_{Res} = 2 + 2 + 2 = 6$. Filled-in ANOVA table:

Source	df	SS	MS	F
Treat	2	38	19	19
Res	6	6	1	–
Total	8	44	–	–

Pairwise $\hat{\tau}(3, 1) = 5$, var $= 1/3 + 1/3 = 2/3, SE \approx 0.82, CI = (3.4, 6.6)$.

5. Crossover Design

Setup

N subjects, each observed in **two periods**. Each subject gets T in one period, C in the other; order chosen by coin flip independently per subject.

$$|W^+| = 2^N.$$

Required assumptions

1. **No carryover**: period-1 treatment doesn't bleed into period 2.
2. **No time/period effect**: the two periods are otherwise exchangeable (no learning, weekday/weekend, adaptation).

Neymanian

Within-subject difference $d_i = Y_i^T - Y_i^C$ (always T minus C, ignoring which came first).

$$\hat{\tau} = \bar{d}, \quad s_D^2 = \frac{1}{N-1} \sum (d_i - \bar{d})^2, \quad \widehat{\text{Var}}(\hat{\tau}) = \frac{s_D^2}{N}.$$

Fisherian

Test statistic: $T_{\text{paired}} = \frac{\bar{d}}{s_D/\sqrt{N}}$.

Under sharp null each d_i 's sign is a coin flip $\Rightarrow$ enumerate all 2^N sign patterns (flip independently per subject), recompute $T_{\text{paired}}^{\text{rand}}$ each time.

$$p = \#\{|T_{\text{paired}}^{\text{rand}}| \geq |T_{\text{paired}}^{\text{obs}}|\} / 2^N.$$

Crossover's unique power

Unit-level effect is estimable: $\hat{\tau}_i = d_i$ is an unbiased estimate of τ_i itself, not just the average τ . This is impossible in CRE (each unit only ever observed once).

Mini example ($N=6$)

$d = \{1.2, 0.8, -0.3, 1.5, 0.6, 1.0\} \Rightarrow \bar{d} = 0.8, \quad s_D^2 = 0.388, \quad T_{\text{paired}} \approx 3.15.$

6. Block Design

Setup

B blocks; block k has $N(k)$ units with $N_t(k)$ treated. Separate CRE within each block.

$$|W^+| = \prod_{k=1}^B \binom{N(k)}{N_t(k)}.$$

Neymanian

Within-block effect: $\hat{\tau}(k) = \bar{Y}_T^{\text{obs}}(k) - \bar{Y}_C^{\text{obs}}(k).$

Overall ATE – population-weighted:

$$\hat{\tau} = \sum_{k=1}^B \lambda_k \hat{\tau}(k), \quad \lambda_k = \frac{N(k)}{N}.$$

Conservative variance:

$$\widehat{\text{Var}}(\hat{\tau}) = \sum_k \lambda_k^2 \left[\frac{s_T^2(k)}{N_t(k)} + \frac{s_C^2(k)}{N_c(k)} \right].$$

Why $N(k)/N$ not $1/B$?

Goal is “average causal effect per person” $\Rightarrow$ weight by the fraction of the population in block k . Using $1/B$ would answer “average across blocks” (ignoring block size).

Fisherian test stat

T_λ :

$$T_\lambda = \sum_k \lambda_k [\bar{Y}_T^{\text{obs}}(k) - \bar{Y}_C^{\text{obs}}(k)].$$

Reference set: **re-randomize within each block**. Size $= \prod_k \binom{N(k)}{N_t(k)}$ (**not** $\binom{N}{N_T}$!). Mixing blocks violates the design.

Effect modification test

Is τ different across blocks? With 2 blocks $k = a, b$:

$$\hat{\Delta} = \hat{\tau}(a) - \hat{\tau}(b), \quad \widehat{\text{Var}} = \sum_{k \in \{a, b\}} \left[\frac{s_T^2(k)}{N_t(k)} + \frac{s_C^2(k)}{N_c(k)} \right].$$

(Variances **add** because blocks are independent; no λ^2 – we're estimating a difference, not a weighted average.)

95% CI $\hat{\Delta} \pm 2SE$; contains 0 $\Leftrightarrow$ no evidence of effect modification.

Purpose of step 7 vs steps 4–6

Steps 4–6 integrate across blocks into one ATE. Step 7 *separates* blocks to ask “does the effect differ?” Both are useful: one gives a summary number, the other diagnoses heterogeneity.

Mini worked example ($B=2$: young/old)

Young: $N(y) = 6, \hat{\tau}(y) = 14 - 10 = 4, s_T^2 = s_C^2 = 4$.
Old: $N(o) = 4, \hat{\tau}(o) = 20 - 17 = 3, s_T^2 = 8, s_C^2 = 2$.
 $\lambda_y = 0.6, \lambda_o = 0.4. \hat{\tau} = 0.6(4) + 0.4(3) = 3.6. \text{Var} = (0.6)^2(4/3 + 4/3) + (0.4)^2(8/2 + 2/2) = 0.96 + 0.80 = 1.76; SE \approx 1.33; \text{CI} = 3.6 \pm 2.65 = (0.95, 6.25).$ Ref. set size $= \binom{6}{3} \binom{4}{2} = 20 \cdot 6 = 120$.

7. Factorial 2^K Design

Lex-order design matrix ($K=3$)

F_1 varies slowest, F_3 varies fastest – binary counting 000 $\rightarrow$ 111.

Run	F_1	F_2	F_3	binary
1	–	–	–	000
2	–	–	+	001
3	–	+	–	010
4	–	+	+	011
5	+	–	–	100
6	+	–	+	101
7	+	+	–	110
8	+	+	+	111

Contrast vectors g_F

Main-effect vectors = columns of the table. Interaction vectors from **Hadamard (elementwise) product**:

$$g_{F_j F_k} = g_{F_j} \odot g_{F_k}, \quad g_{F_1 F_2 F_3} = g_{F_1} \odot g_{F_2} \odot g_{F_3}.$$

Rule: **same sign** $\rightarrow +$, **different sign** $\rightarrow -$.

Sanity: every non-grand-mean g_F has 2^{K-1} pluses and 2^{K-1} minuses; distinct g_F 's are orthogonal.

Factor effect estimator

With $Q = 2^K$ cells, observed cell means $\bar{Y}^{\text{obs}}$ length Q :

$$\hat{\tau}_F = \frac{1}{2^{K-1}} g_F^\top \bar{Y}^{\text{obs}}.$$

Interpretation of $\hat{\tau}_{F_j}$: average change in response when F_j flips from $-$ to $+$, averaged over all combos of other factors.

Neymanian variance (equal n per cell)

Pool within-cell variances: $\hat{S}^2 = \frac{1}{2^K} \sum_w s_w^2$.

$$\widehat{\text{Var}}(\hat{\tau}_F) = \frac{2^K}{N} \hat{S}^2.$$

Same formula for every effect (main or interaction). 95% CI: $\hat{\tau}_F \pm 2SE$.

Fisherian for factorial

Test statistic: $\hat{\tau}_F$ itself.

Under sharp null ($\tau_{iF} = 0 \forall i, F$), each unit's 2^K potential outcomes are identical. Y^{obs} fixed; re-randomize N units into Q cells (n each). Total perms $= N!/(n!)^Q$ – usually huge $\Rightarrow$ Monte Carlo. Recompute $\hat{\tau}_F^{\text{rand}}$ each draw; two-sided $p = \#\{|\hat{\tau}_F^{\text{rand}}| \geq |\hat{\tau}_F^{\text{obs}}|\} / B$.

Practical meaning of g_F

g_F is a **recipe**: “+” = add that cell, “–” = subtract it. $g_{F_1 F_2}^\top \bar{Y} / 2^{K-1}$ is exactly the “difference of differences” that defines interaction – just mechanized. Orthogonality $\Rightarrow$ each effect estimated independently (no contamination).

Mini worked example ($K=3$, baking)

Cell means in lex order $\bar{Y}^{\text{obs}} = (4, 6, 5, 9, 5, 7, 8, 14)$. Divisor $= 2^{K-1} = 4$. $\hat{\tau}_{F_1} = (-4 - 6 - 5 - 9 + 5 + 7 + 8 + 14)/4 = 10/4 = 2.5$. $\hat{\tau}_{F_2} = (-4 - 6 + 5 + 9 - 5 - 7 + 8 + 14)/4 = 14/4 = 3.5$. $\hat{\tau}_{F_3} = (-4 + 6 - 5 + 9 - 5 + 7 - 8 + 14)/4 = 14/4 = 3.5$. $\hat{\tau}_{F_1 F_2} = (+4 + 6 - 5 - 9 - 5 - 7 + 8 + 14)/4 = 6/4 = 1.5$ (interaction). With $n=2$ per cell, $s_w^2 = 2 \Rightarrow \hat{S}^2 = 2$; $\widehat{\text{Var}}(\hat{\tau}_F) = \frac{8}{16} \cdot 2 = 1 \Rightarrow SE = 1$; each 95% CI $= \hat{\tau}_F \pm 2$.

8. Fractional Factorial

Why

Full 2^K needs 2^K runs – often too expensive. A 2^{K-p} fraction uses only 2^{K-p} runs at the cost of aliasing.

Half-fraction 2^{4-1} via $D = ABC$

Start with 2^3 lex table on A, B, C . Set column $D =$ elementwise product $A \odot B \odot C$.

#	A	B	C	D=ABC
1	-	-	-	-
2	-	-	+	+
3	-	+	-	+
4	-	+	+	-
5	+	-	-	+
6	+	-	+	-
7	+	+	-	-
8	+	+	+	+

Defining relation & aliasing

The generator $D = ABC \Leftrightarrow$ defining relation $I = ABCD$.
Multiply both sides by any effect X (using $X^2 = I$, i.e. same column squared = all +1):

- $A \cdot I = A \cdot ABCD \Rightarrow A \equiv BCD$
- $AB \cdot I = AB \cdot ABCD \Rightarrow AB \equiv CD$

Aliased effects share the same contrast column $\Rightarrow$ indistinguishable from 8 runs alone.

Full alias table for $I = ABCD$

Clear pair		Clear pair	
I	ABCD	D	ABC
A	BCD	AB	CD
B	ACD	AC	BD
C	ABD	AD	BC

Resolution

Shortest word length in defining relation = **resolution**. For $I = ABCD$, length 4 $\Rightarrow$ **Resolution IV**.

- Res III: main aliased with 2fi ($\rightarrow$ risky)
- Res IV: main clear of 2fi, but 2fi aliased with 2fi
- Res V: main & 2fi clear, 2fi aliased with 3fi (good!)

Estimating effects in a 2^{K-1}

Use the 2^{K-1} lex table as if it were a full 2^{K-1} . Effect formula:

$$\hat{\tau}_F = \frac{1}{2^{(K-1)-1}} g_F^\top \bar{Y}^{obs} = \frac{g_F^\top \bar{Y}^{obs}}{2^{K-2}}.$$

What you actually estimate is $\tau_F + \tau_{F'}$ where F' is F 's alias. If higher-order alias is negligible, $\hat{\tau}_F \approx \tau_F$.

Worked example

$\bar{Y}^{obs} = (20, 25, 24, 27, 23, 30, 28, 33)$ in the order above.
 $g_A^\top \bar{Y} = -20 - 25 - 24 - 27 + 23 + 30 + 28 + 33 = 18$.
 $\hat{\tau}_A = 18/4 = 4.5$ (with $K = 3$ for 8-run fraction). Strictly estimating $\tau_A + \tau_{BCD}$; assuming $\tau_{BCD} \approx 0$, main effect of $A \approx 4.5$.

Design	Test stat T	Reference set
CRE	$T_{dm} = \bar{Y}_T - \bar{Y}_C$	$\binom{N}{N_T}$ label perms
Matched Pair	$T = d$	2^P sign flips within pairs
Multi-Trt CRE	$F = \frac{MSTr}{MSRes}$	$\frac{N!}{\prod N_j!}$ perms (one-sided)
Crossover	$T_{paired} = \frac{\bar{d}}{s_D/\sqrt{N}}$	2^N per-subject sign flips
Block	$T_\lambda = \sum_k \lambda_k \Delta_k$	$\prod_k \binom{N(k)}{N_t(k)}$ within-block
Factorial 2^K	$\hat{\tau}_F$ itself	$N!/(n!)2^K \rightarrow$ Monte Carlo

Generic Fisherian recipe (memorize!)

1. State sharp null $H_0 : Y_i(1) = Y_i(0) \forall i$ (or all factor effects = 0).
2. Treat the N observed numbers as **fixed**.
3. Enumerate/sample the **design-specific** reference set of assignments.
4. For each, recompute T^{rand} .
5. $p = \#\{|T^{rand}| \geq |T^{obs}|\} / |W^+|$ (two-sided unless F).

10. Symbol Glossary (Quick Examples)

General

$W_i \in \{0, 1\}$ – treatment indicator for unit i (e.g. $W_3 = 1$ means plot 3 got new fertilizer)

$Y_i(w)$ – potential outcome under w (e.g. $Y_4(1) = 34$ is plot 4's yield if treated)

$Y_i^{obs} = Y_i(W_i)$ – the one outcome we actually see (e.g. $Y_4^{obs} = 34$)

$\tau_i = Y_i(1) - Y_i(0)$, unit-level causal effect (e.g. if $Y_1(1) = 30, Y_1(0) = 25$, then $\tau_1 = 5$)

$\tau = \frac{1}{N} \sum \tau_i$, finite-population ATE (e.g. $\tau = 5$ avg across 6 plots)

$\hat{\tau}$ – estimator of τ from one realized W (e.g. $\hat{\tau} = 9.67$ here even though $\tau = 5$)

$S_\tau^2 = \frac{1}{N-1} \sum (\tau_i - \tau)^2$, finite-pop var of τ_i (e.g. **unobservable** – drop in CRE variance)

s_j^2 – sample variance in arm j , $\div (N_j - 1)$ (e.g. $s_1^2 = 4$ for $\{30, 32, 34\}$)

Matched Pair / Crossover

d_p or d_i – within-pair/within-subject diff, $T - C$ (e.g. $d_1 = 8 - 5 = 3$)

$\bar{d}$ – mean of differences (e.g. $\bar{d} = 3$ for $d = \{3, 1, 5, 3\}$)

s_d^2, s_D^2 – sample var of differences, $\div (P - 1)$ or $(N - 1)$ (e.g. $s_d^2 = 8/3$)

T_{paired} – paired t -stat $= \bar{d} / (s_D / \sqrt{N})$ (e.g. $T_{paired} \approx 3.15$ if $\bar{d} = 0.8, s_D = 0.62, N = 6$)

Multi-Treatment / ANOVA

$\bar{Y}_j$ – mean of arm j (e.g. $\bar{Y}_A = 4$ for $\{3, 5\}$)

$\bar{Y}$ – grand mean $= \frac{1}{N} \sum Y_i^{obs}$ (e.g. $\bar{Y} = 22/3$ for 3 arms means $\{4, 7, 11\}$)

$SS_{Treat} = \sum_j N_j (\bar{Y}_j - \bar{Y})^2$ (e.g. measures between-arm separation)

$SS_{Res} = \sum_{j,i} (Y_i - \bar{Y}_j)^2$ (e.g. within-arm noise)

$F = MST_{Treat} / MS_{Res}$ (e.g. $F \approx 1$ under H_0 ; large F rejects)

Block

$N(k)$ – size of block k (e.g. $N(y) = 6$ young)

$\lambda_k = N(k)/N$, population weight (e.g. $\lambda_y = 0.6$ out of $N = 10$)

$\hat{\tau}(k)$ – within-block effect $\bar{Y}_T(k) - \bar{Y}_C(k)$ (e.g. $\hat{\tau}(y) = 4$)

$T_\lambda = \sum_k \lambda_k \hat{\tau}(k)$ (e.g. $T_\lambda = 0.6(4) + 0.4(3) = 3.6$)

Factorial

K – # two-level factors (e.g. $K = 3$ for temp $\times$ time $\times$ flour)

g_F – contrast vector for effect F , length 2^K (e.g. $g_{F_1} = (-, -, -, -, +, +, +, +)$)

$g_{F_j F_k} = g_{F_j} \odot g_{F_k}$ (elementwise product) (e.g. $g_{F_1 F_2} = (+, +, -, -, -, -, +, +)$)

$\hat{\tau}_F = \frac{1}{2^{K-1}} g_F^\top \bar{Y}^{obs}$ (e.g. $\hat{\tau}_{F_1} = 10/4 = 2.5$)

$\hat{S}^2 = \frac{1}{2^K} \sum_w s_w^2$, pooled within-cell var (e.g. $\hat{S}^2 = 2$ if all cell vars are 2)

Fractional Factorial

I – identity / grand-mean column (e.g. all +1's; $X \cdot I = X$)

$D = ABC$ – generator for 2^{4-1} half-fraction (e.g. builds column D from A, B, C)

$I = ABCD$ – defining relation (e.g. multiply by X to find X 's alias)

$A \equiv BCD$ – alias pair (same contrast column) (e.g. can't separate τ_A from τ_{BCD} with 8 runs)

Resolution – min word length in defining relation (e.g. $ABCD \Rightarrow$ Res IV)

11. Conceptual Q&A (High-Yield)

Q1. Why is CRE variance “conservative” but Matched Pair “exact”?

CRE: true variance contains S_τ^2/N , which requires both potential outcomes per unit – unobservable. We drop it, giving an upper bound.

MP: d_p is directly observable for each pair; no hidden term. s_d^2/P is unbiased for $\text{Var}(\hat{\tau})$, no dropping needed.

Q2. Why 95% CI uses ± 2 , not ± 1.96 ?

STAT 140 convention: 2 is the rule-of-thumb for a 95% interval under approximate normality. Using ± 2 is slightly more conservative (wider) than ± 1.96 , so you never under-cover. Always ± 2 on exams unless problem says otherwise.

9. Fisherian T 's by Design

Q3. Why is Block's weight $N(k)/N$ rather than $1/B$?

Target is "avg causal effect per *person* in the population". Blocks with more people should count more. $1/B$ answers "avg across blocks" (ignores size) – different estimand.

Q4. Why is Block's reference set $\prod_k \binom{N(k)}{N_t(k)}$, not $\binom{N}{N_T}$?

The design re-randomizes *only within each block*. Using $\binom{N}{N_T}$ allows swapping units between blocks – violates design. Fewer legal assignments $\Rightarrow$ tighter inference.

Q5. Why can Crossover estimate τ_i but CRE cannot?

Crossover observes each unit under *both* conditions (across the two periods). d_i directly gives τ_i . In CRE each unit is observed under only one condition, so τ_i is always missing one potential outcome.

Q6. Matched Pair reference set: 2^P or $\binom{N}{N/2}$?

2^P . The design restricts randomization to *within* pairs. $\binom{N}{N/2}$ would allow cross-pair swaps $\Rightarrow$ wrong null distribution, wrong p -value.

Q7. In Multi-Treatment CRE, why gate with F before pairwise?

$\binom{J}{2}$ pairwise tests inflate family-wise error to $\approx 1 - (1 - \alpha)^{\binom{J}{2}}$ (≈ 0.14 at $J = 3, \alpha = 0.05$). The F -test provides one overall check; only if F rejects do we look at pairs.

Q8. Why does Factorial use $\hat{\tau}_F$ *itself* as the Fisherian T ?

Because the design is already symmetric – every g_F has equal $+$'s and $-$'s. Under sharp null, $E[\hat{\tau}_F] = 0$ and its randomization distribution is symmetric about 0. No need for a studentized ratio.

Q9. What does "Resolution IV" mean in fractional factorial?

Shortest word in the defining relation has length 4. Consequence: main effects are *clear* of all 2-factor interactions, but 2fi's alias with each other. Res III = main aliased with 2fi (bad). Res V = main and 2fi both clear of each other (best; needs more runs).

Q10. When do we need Monte Carlo vs full enumeration?

Enumerate when $|W^+|$ is small: MP ($2^5 = 32$), small block ($\leq 10^3$), small CRE ($\binom{8}{4} = 70$). MC when too big: CRE with $N = 30$, Factorial reshuffle ($N!/(n!)^Q$). Standard $B = 10,000$.

Q11. "Effect modification" vs "overall ATE" – are they competing?

No, complementary. Overall ATE (integrated via λ_k) gives population summary. Effect modification (difference $\hat{\tau}(a) - \hat{\tau}(b)$) diagnoses whether the effect *depends on block*. A significant overall ATE + significant effect modification means "it works on average but more strongly for some subgroup".

Q12. TI-Nspire CX quick variance

For $\{3, 1, 5, 3\}$: use `variance({3,1,5,3})` in a calc page. Returns sample variance (divides by $n - 1$) directly. Do **not** square the stdev manually – risk of rounding.

Q13. What does $\hat{\tau}_{F_1 F_2}$ mean; why is $\hat{\tau}_{F_1}$ tricky if $\hat{\tau}_{F_1 F_2}$ is large?

$\hat{\tau}_{F_1 F_2}$ is the **interaction**: how much the effect of F_1 *changes* when F_2 flips $- \rightarrow +$ (a "difference of differences"; equivalently, half of [effect of F_1 at $F_2=+$] – [effect of F_1 at $F_2=-$]).

If $\hat{\tau}_{F_1 F_2}$ is large, $\hat{\tau}_{F_1}$ is only an *average* over the two levels of F_2 – the real effect of F_1 depends on F_2 , so reporting $\hat{\tau}_{F_1}$ alone hides the true pattern (could even flip sign in one F_2 cell).

Final cheat summary

Design	$ W^+ $	Variance nature	Fisher T
CRE	$\binom{N}{N_T}$	Conservative (drops S_τ^2)	$\bar{Y}_T - \bar{Y}_C$
MP	2^P	Exact s_d^2/P	$\bar{d}$
Multi-Trt	$N! / \prod N_j!$	Conservative	F
Crossover	2^N	Based on s_D^2/N	T_{paired}
Block	$\prod_k \binom{N(k)}{N_t(k)}$	Conservative	T_λ
Fact. 2^K	$N!/(n!)^{2^K}$ (MC)	$\frac{2^K}{N} \hat{S}^2$	$\hat{\tau}_F$
Frac. 2^{K-p}	2^{K-p} runs	same form, $K \rightarrow K - p$	$\hat{\tau}_F$ (aliased)

Good luck, Casval! Read the design, write the table, then plug numbers.

12. Exam-Taking Flow

- Identify the design** first (CRE? MP? Crossover? Block? Factorial? Fractional?). Everything flows from this.
- If 2^K or $2^{K-p} \rightarrow$ **write the lex-order table** first, then g_F , then plug numbers.
- If fractional $\rightarrow$ **write defining relation** $I = \dots$, then list alias pairs, then compute as if 2^{K-p} full factorial.
- If Fisherian $\rightarrow$ **state sharp null, fix observed numbers, identify reference set size, pick T** , then enumerate or MC.
- If Neymanian $\rightarrow \hat{\tau}$, sample variances, $\widehat{\text{Var}}, SE, \hat{\tau} \pm 2SE$.
- Units check**: df, divisors, 2^{K-1} vs 2^K , $N_t(k)$ vs $N(k)$ – most errors here.
- For CIs / variances: ask yourself *conservative or exact?* If dropping $S_\tau^2 \rightarrow$ conservative.